COMSC 78 **Study Guide for Test #1**

The test will cover all of the sections in Chapter 1 in the course text. The proctored portion of the exam will consist of multiple-choice questions and short answers questions (including short programs). You will be asked to reproduce some functions drawn from sections 1.6 and 1.7. It will be worth 50 points.

1. Describe, succinctly, what computing is about.
2. What do computer programs consist of?
3. What is an interpreter?
4. How do programming languages provide for combining simple ideas to form more complex ideas.
5. Programming deals with two kinds of elements. What are they?
6. What are the three principal advantages that function notation has over infix notation?
7. What is the difference between Pure and non-Pure functions?
8. How does the textbook define an “environment”, and why is this important?
9. What is an environment diagram?
10. What is a function signature?
11. What is a “good” function according to the textbook?
12. What is a docstring? Why is it useful?
13. What is a default value and when is it used in Python programs?
14. What is the purpose of control statements?
15. What is a conditional statement?
16. Describe short-circuiting.
17. What are the key steps in the execution of a while loop?
18. What are assertions?
19. What is a higher-order function?
20. Why are functions considered to be fundamentally important in writing programs?
21. What is meant by lexical scope?
22. What are the advantages of lexical scoping in Python?
23. What does it mean for a programming language to have first-class status?
24. What is a recursive function?
25. Can any recursive solution be replaced by an iterative one?
26. What is mutual recursion? Give an example. Can we replace every mutual recursion with simple recursion?
27. Be prepared to write several recursive functions.

Make sure to manage your time well enough to get to each of these topics during the week preceding the exam. Re-read all of the sections in the text, paying particular attention to the programs in sections 1.6 and 1.7.

The second part of the exam will have you write a Python program which you will upload to Canvas for Test #1. It will be worth 25 points.

Best of luck on the exam.